EXERCISE — Run Phase

Your target is the **WannaCry ransomware Bitcoin wallets** — fully documented, publicly attributed to North Korean Lazarus Group, perfect for practicing transaction graph analysis.

Three known WannaCry payment addresses are publicly documented:

```
12t9YDPgwueZ9NyMgw519p7AA8isjr6SMw
115p7UMMngoj1pMvkpHijcRdfJNXj6LrLn
13AM4VW2dhxYgXeQepoHkHSQuy6NgaEb94
```

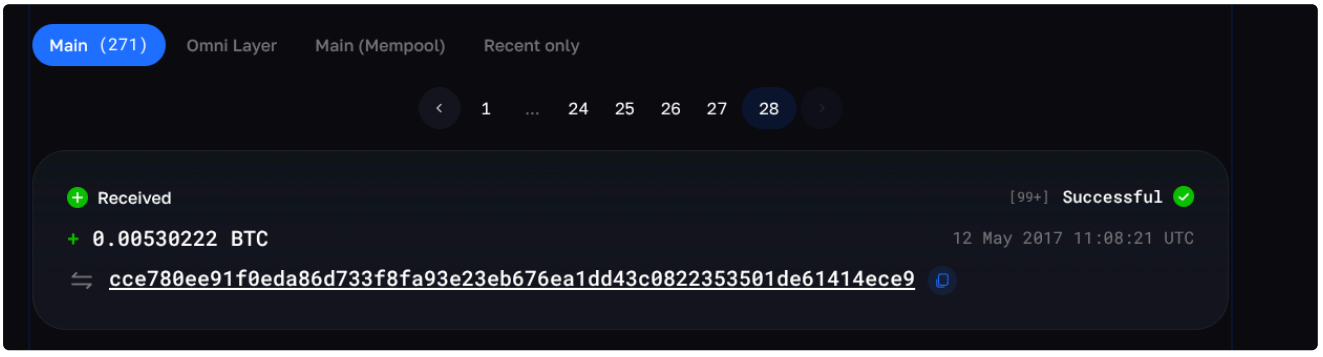
Using only free public tools — Blockchair, OXT, or WalletExplorer — investigate all three wallets and document:

- Total funds received across all three
- When did activity start and stop
- Did funds move anywhere after collection
- Can you identify any exchange attribution
- What does the transaction pattern tell you about the operator

Write your full analyst note with confidence ratings.

When did activity start and stop?

Confidence: High



i'll consider the "start" of the operation the earliest time one of the three wallets recieved its BTC, in this case 12 May 2017 11AM UTC

while the "Stop" will be the time of the last sent amount of BTC by the MAIN three not a puppet wallet, because the main ones are directly connected to the malware.
in this case:

ID: 35e5-d8a7 8/03/2017, 05:41:34	From 12t9-6SMw To 16df-Z8Vy	-9.03851401 BTC • -$736,910 Fee 1.1M Sats • $860.21
ID: 4098-1698 8/03/2017, 05:28:20	From 12t9-6SMw To 2 Outputs	-8.73261636 BTC • -$711,971 Fee 505.1K Sats • $411.82

Total funds received across all three

Confidence: high

First One:

HASH

12t9YDPgwueZ9NyMgw519p7AA8isjr6SMw

MAIN BALANCE

+ 1.92242576 BTC · 154,974.29 USD

RECENT TRANSACTION

+ 0.0006 BTC · 36.45 USD

03 Oct 2024 11:37:09 UTC

View in history

The total of the balance

Main (360)

1 2 3 4 5 ... 37

Received [99+] Successful

+ 0.0006 BTC · 36.45 USD 03 Oct 2024 11:37:09 UTC

81e15438df4d1e242afc732046ec05f6d8e3e36f9fe167477305f0e09e34e543

Received [99+] Successful

+ 0.00012615 BTC · 8.68 USD 03 Jun 2024 00:07:00 UTC

d99660720b55ce7a9d34d8aaa5c60013fd1fc162807980154107c463a151a382

Received [99+] Successful

+ 0.00010201 BTC · 8.22 USD 06 Mar 2023 07:33:42 UTC

248d507bf9a07130bec3f4941e2e045e5b05939983b249c733ceb3f386d45362

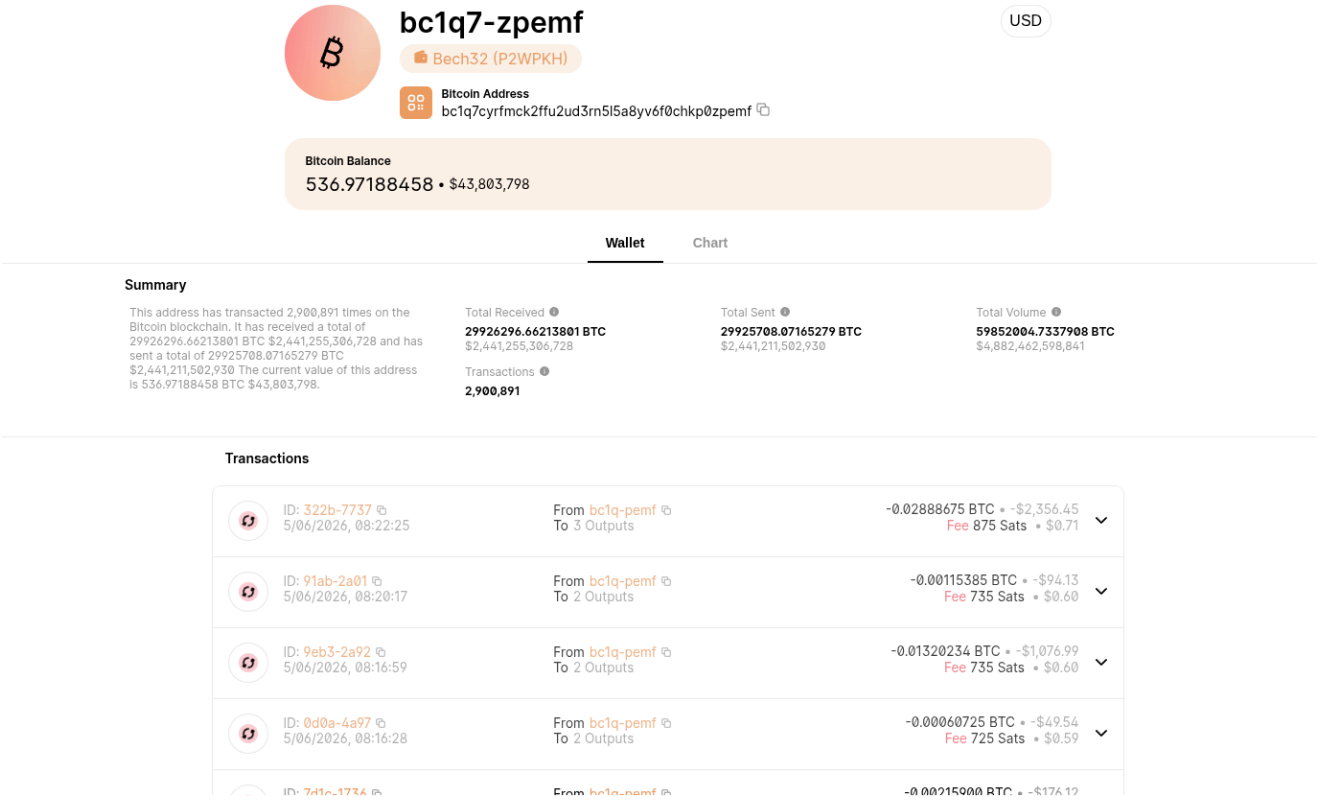
Received [99+] Successful

+ 0.00243992 BTC · 196.69 USD 10 Feb 2023 21:57:25 UTC

8ae38558912466ec56a9948c6aa0ffbb3ece2d523dca83288f07cac1480a1bb5

history of transactions

Looking deeper into the *Last* Recieved transaction we can find out that the wallet that sent the amount of bitcoins is a quite interesting one:



HASH

bc1q7cyrfmck2ffu2ud3rn5l5a8yv6f0chkp0zpemf

MAIN BALANCE

+ 588.89475012 BTC · 48,055,996.40 USD

RECENT TRANSACTION

- 12.91793102 BTC · 1,054,151.09 USD

Pending confirmation since 2 minutes ago

View in history

History

Recent only | Main (6,010,725) | Main (Mempool)

Sent — Pending
- 12.91793102 BTC · 1,054,151.09 USD — 2 minutes ago
2cb27d27eae321140213a22a3c1f13c9b5fd3e4dea8983607e75f7cd07b018be

Received — Pending
+ 12.82419924 BTC · 1,046,502.23 USD — 2 minutes ago
2cb27d27eae321140213a22a3c1f13c9b5fd3e4dea8983607e75f7cd07b018be

Received — Pending
+ 12.92613782 BTC · 1,054,820.80 USD — 3 minutes ago
5563e66b69a3ca2835f79f26dc796b6b5e619cca20bdabc193629acb8bc79242

Sent — Pending
- 12.92987188 BTC · 1,055,125.51 USD — 3 minutes ago
5563e66b69a3ca2835f79f26dc796b6b5e619cca20bdabc193629acb8bc79242

The suspicious thing is the amount owned by the wallet since its not common to own 588+ bitcoins, but also, the number of transactions it has (6m+)

- The last 4 pages are minutes ago, not days and all has great amounts of bitcoin in it

Second One:

HASH

115p7UMMngoj1pMvkpHijcRdfJNXj6LrLn

MAIN BALANCE

+ 0.46702392 BTC · 38,057.78 USD

RECENT TRANSACTION

+ 0.0372 BTC · 3,031.42 USD

14 Nov 2020 07:08:23 UTC

View in history

The total of the balance

Main (234)

1 2 3 4 5 ... 24

Received [99+] Successful

+ 0.0372 BTC · 3,031.42 USD 14 Nov 2020 07:08:23 UTC

14449446275da0bf11825d14733fcc28f7264f8a2c3a506752f92fddb8e1aa16

Received [99+] Successful

+ 0.0002783 BTC · 22.67 USD 07 Mar 2020 23:12:48 UTC

4f9c82c08d85f294a47b1b966fa9107dc690ffc5b314a7379097bfdd9483d3ce

Received [99+] Successful

+ 0.03011781 BTC · 2,454.30 USD 01 Sep 2019 07:06:29 UTC

5042e1347182ce055d85c9160b396125d434c6e540ce3c5e5a71b06a91fb4529

Received [99+] Successful

+ 0.07766226 BTC · 6,328.69 USD 29 Dec 2018 11:52:50 UTC

c3350dc0e1023532b294edd77b9b4d70c61133358d0dd47be6d15a44491f8053

history of transactions

Third One:

HASH

13AM4VW2dhxYgXeQepoHkHSQuy6NgaEb94

MAIN BALANCE

+ 0.32943048 BTC · 26,805.75 USD

RECENT TRANSACTION

+ 0.0005 BTC · 30.16 USD

10 Oct 2024 17:32:17 UTC

View in history

The total of the balance

Main (271)

1 2 3 4 5 ... 28

Received [99+] Successful
+ 0.0005 BTC · 30.16 USD 10 Oct 2024 17:32:17 UTC
88fbcc13e7b587dfad660b912aaaeedec72e261f582a78e2a4f1a06b4159ac1e

Received [99+] Successful
+ 0.0005 BTC · 30.31 USD 09 Oct 2024 15:58:14 UTC
12ea399fb5073f1b64b3b93caa35ad1d77e17f564b9e6ff3a4dcfc9396f7f6b7

Received [99+] Successful
+ 0.0077 BTC · 626.54 USD 27 May 2021 13:14:31 UTC
9ff0abcedf98c3cf1eee581f44d8cdc66d8c0199f28134e161b5f8b6f507a291

Received [99+] Successful
+ 0.001 BTC · 81.37 USD 30 Aug 2020 14:35:57 UTC
ed67062ecc8a8e552f82618d4d842c1bd6ac835ff26febdf8f4141b84ccb69b0

history of transactions

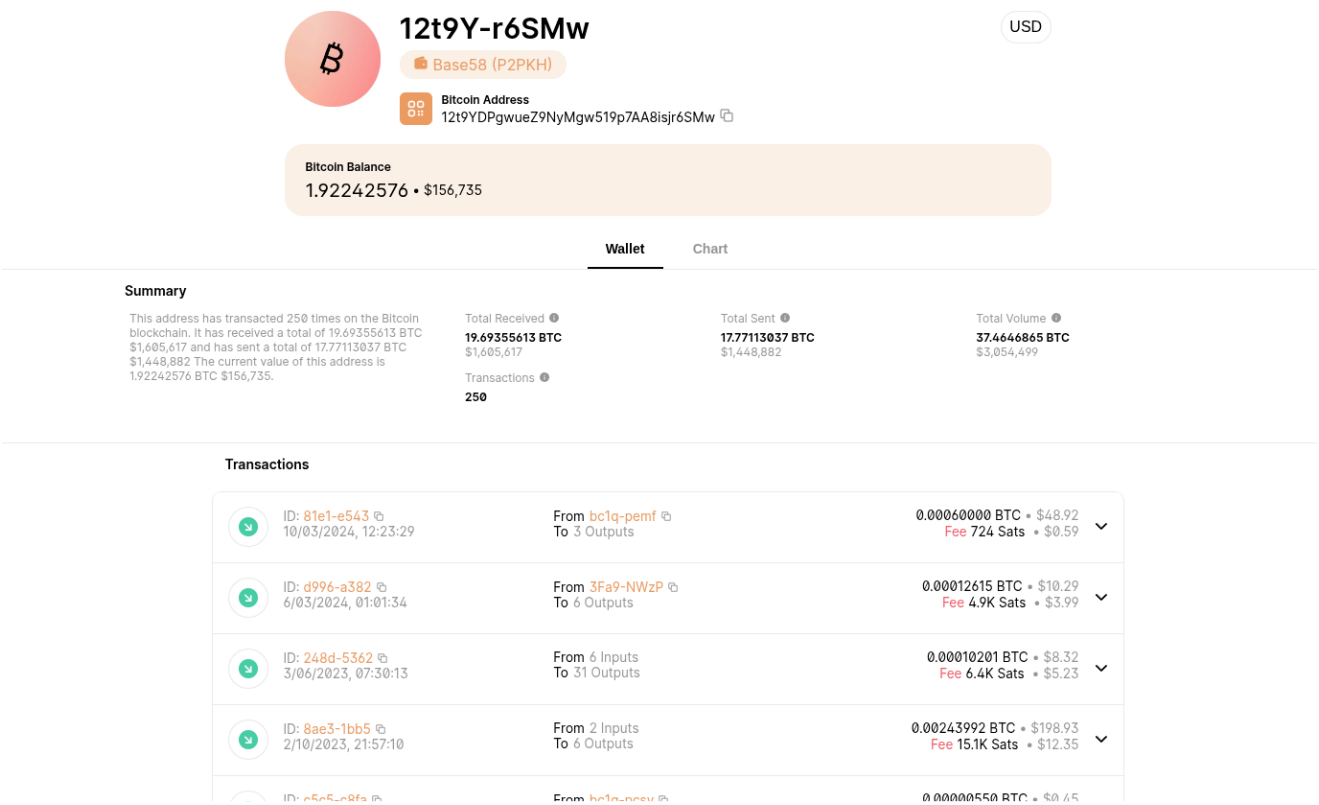
So what's the total across the three wallets?

Total cash inflows from all three sources is: 219,837.82 USD

Did funds move anywhere after collection?

Confidence: high

First Wallet Transactions Overview

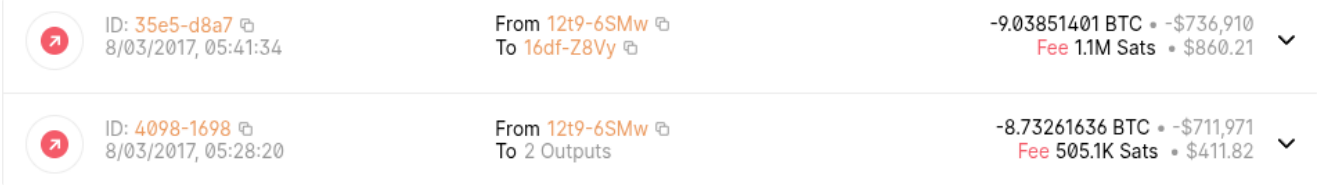


The first wallet has recieved a total of 19.69355613 BTC and has sent a total of 17.77113037 BTC, knowing this we can assume the BTC not sent were recieved after the operation got disrupted.

This address has transacted 250 times on the Bitcoin blockchain. It has received a total of 19.69355613 BTC $1,602,627 and has sent a total of 17.77113037 BTC $1,446,183 The current value of this address is 1.92242576 BTC $156,443.

Where did it sent its BTC?

This is the last time this wallet has sent an amount of BTC



Both on the same day, in distance of minutes between each other, but one has 2 outputs, ill start from there:

From		To	
1	12t9YDPgwueZ9NyMgw519p7AA8isjr6SMw 0.17124673 BTC • $13,961.76	1	1JC41YHmjKEcW1rLH6pmMWEFHkoNwSmhnC 0.01227173 BTC • $1,000.52
2	12t9YDPgwueZ9NyMgw519p7AA8isjr6SMw 0.18010230 BTC • $14,683.76	2	1FQQ86tMuvhQ4Ruyqgbb8j7iaNfUZ69gpY 8.71529348 BTC • $710,558
3	12t9YDPgwueZ9NyMgw519p7AA8isjr6SMw 0.16656716 BTC • $13,580.24		
4	12t9YDPgwueZ9NyMgw519p7AA8isjr6SMw 0.17030000 BTC • $13,884.58		
5	12t9YDPgwueZ9NyMgw519p7AA8isjr6SMw 0.16900000 BTC • $13,778.59		
6	12t9YDPgwueZ9NyMgw519p7AA8isjr6SMw 0.18040000 BTC • $14,708.03		
7	12t9YDPgwueZ9NyMgw519p7AA8isjr6SMw 0.17887884 BTC • $14,584.01		
8	12t9YDPgwueZ9NyMgw519p7AA8isjr6SMw 0.33600000 BTC • $27,394.11		
9	12t9YDPgwueZ9NyMgw519p7AA8isjr6SMw 0.32000000 BTC • $26,089.63		
10	12t9YDPgwueZ9NyMgw519p7AA8isjr6SMw 0.14860000 BTC • $12,115.37		

Load 26 More

Here's the first wallet (1JC41YHmjKEcW1rLH6pmMWEFHkoNwSmhnC):

HASH

1JC41YHmjKEcW1rLH6pmMWEFHkoNwSmhnC

MAIN BALANCE

+ 0 BTC · 0.00 USD

RECENT TRANSACTION

- 0.01227173 BTC · 1,000.24 USD

03 Aug 2017 06:45:42 UTC

View in history

Main (2)

Sent [99+] Successful

- 0.01227173 BTC · 1,000.24 USD 03 Aug 2017 06:45:42 UTC

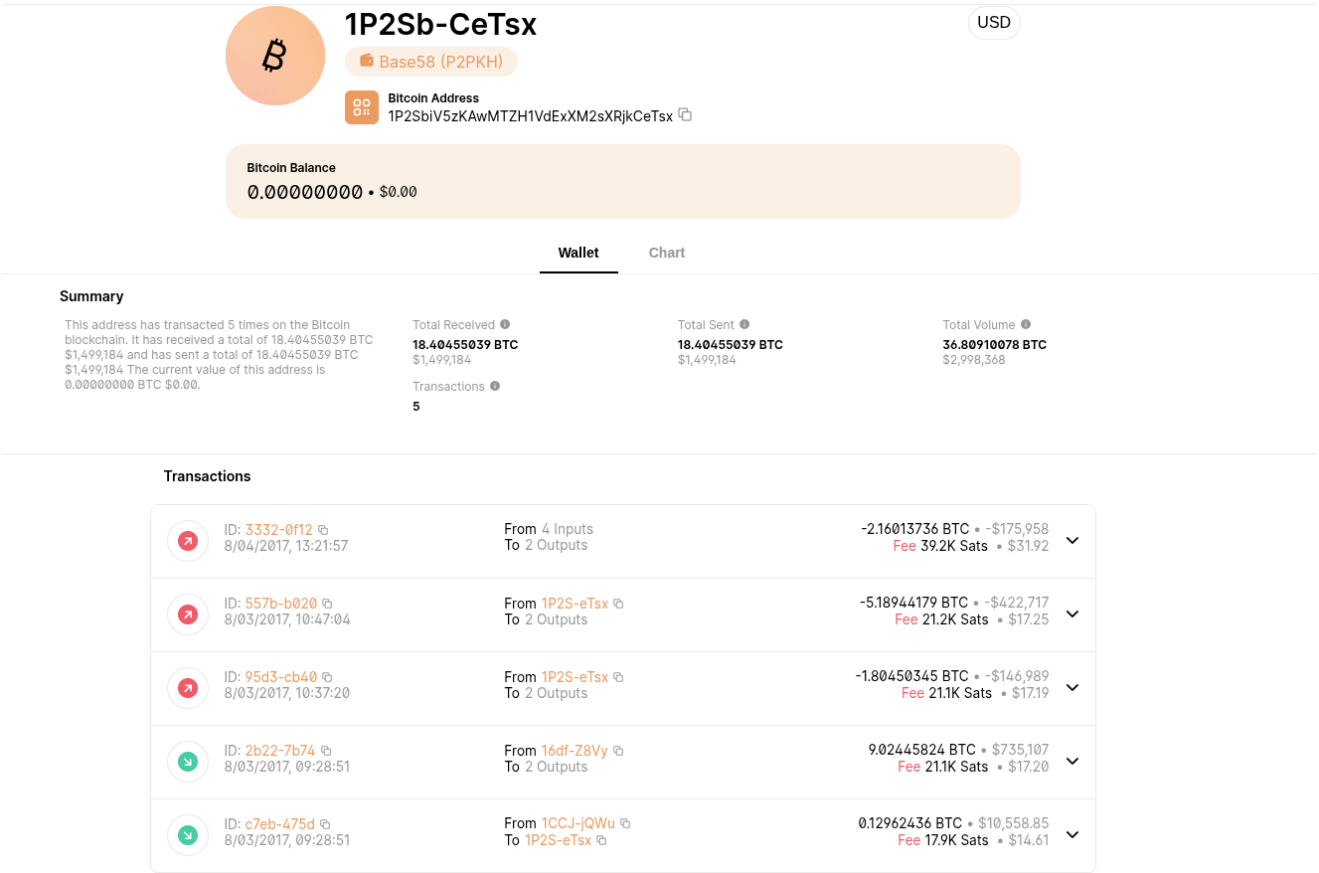
c03e48ad9fc778170c86542c0414a89052b21679a3576121ca6b1c2d340f1e22

Received [99+] Successful

+ 0.01227173 BTC · 1,000.24 USD 03 Aug 2017 04:28:20 UTC

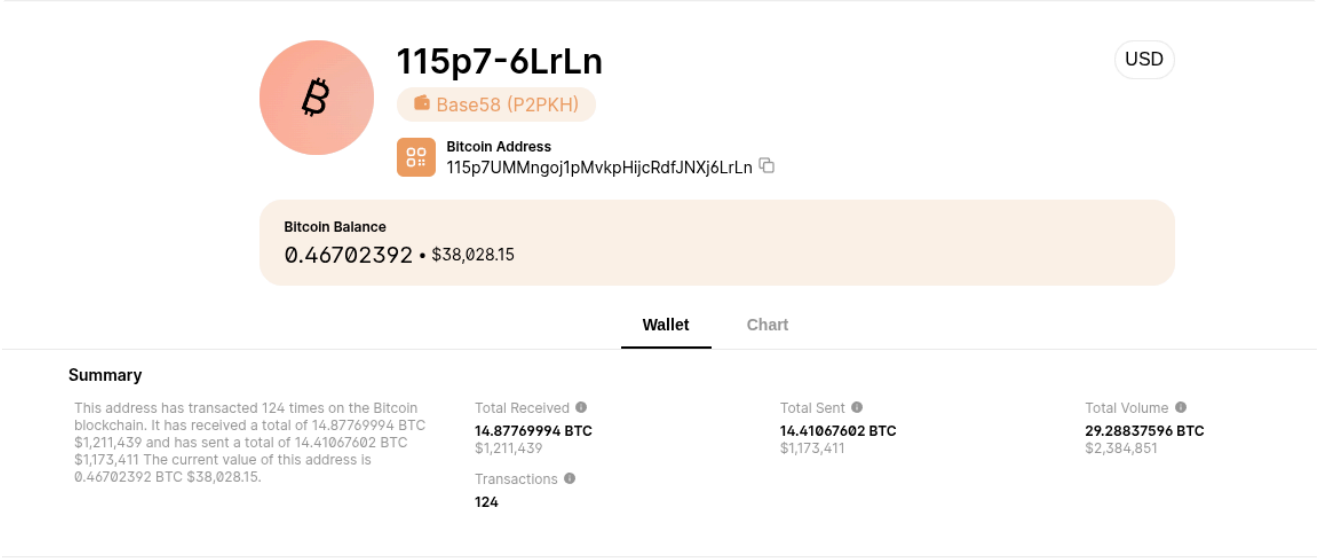
409803bb5e124fd028c0482027c7722e84ce55b78204b279d3a44aba5e7c1698

Knowing that this are his only transaction i can assume its a "puppet wallet" and it sends those BTC to another puppet in a long loop until it finally reaches the actual wallet, ill follow the loop until i find an actual candidate



This puppet is pretty different from the previous one, because it sends the BTC into different wallets, i wont follow this leash since its out-of-scope my orders.

Second wallet transactions overview



The first wallet has recieved a total of 14.87769994 BTC and has sent a total of 14.41067602 BTC, knowing this we can assume the BTC not sent were recieved after the operation got disrupted.

This address has transacted 124 times on the Bitcoin blockchain. It has received a total of 14.87769994 BTC $1,211,439 and has sent a total of 14.41067602 BTC $1,173,411 The current value of this address is 0.46702392 BTC $38,028.15.

Where did it sent its BTC?

This is the last time this wallet has sent an amount of BTC:

ID: d178-9ab8 8/03/2017, 05:41:34	From 115p-LrLn To 2 Outputs	-7.34128314 BTC • -$597,775 Fee 574.3K Sats • $467.60
ID: ef0a-cf2f 8/03/2017, 05:41:34	From 115p-LrLn To 14Y8-VQif	-7.06939288 BTC • -$575,636 Fee 985.9K Sats • $802.81

From this i've started to see an actual pattern in their cycle of transactions:

- 1st: As the ransomware works, they recieve BTC to the three wallets
- 2st: Each wallets before the operation was disrupted has sent all the current BTC into 2 transaction, one has 2 outputs each time while the other just one
- 3st: The single one starts a puppet wallet loop to distract the researcher, and making him waste time following this loop as time goes by

This wallets does the same things as the first, just different wallets it send the BTC to, ill move to the third without wasting more time.

Third one

13AM4-aEb94

USD

Base58 (P2PKH)

Bitcoin Address
13AM4VW2dhxYgXeQepoHkHSQuy6NgaEb94

Bitcoin Balance
0.32943048 • $26,848.19

Wallet Chart

Summary

This address has transacted 145 times on the Bitcoin blockchain. It has received a total of 20.07453352 BTC $1,636,050 and has sent a total of 19.74510304 BTC $1,609,202 The current value of this address is 0.32943048 BTC $26,848.19.

Total Received	Total Sent	Total Volume
20.07453352 BTC $1,636,050	19.74510304 BTC $1,609,202	39.81963656 BTC $3,245,252

Transactions
145

Last transaction sent by this wallet:

ID: 8def-93a1 8/03/2017, 05:28:20	From 13AM-Eb94 To 1M1C-HJbb	-10.06868926 BTC • -$820,586 Fee 1.1M Sats • $871.15
ID: a028-cc4c 8/03/2017, 04:39:15	From 13AM-Eb94 To 2 Outputs	-9.67641378 BTC • -$788,616 Fee 1.1M Sats • $871.07

This confirms my theory from earlier:

```
From this i've started to see an actual pattern in their cycle of
transactions:

- 1st: As the ransomware works, they recieve BTC to the three wallets
- 2st: Each wallets before the operation was disrupted has sent all the
current BTC into 2 transaction, one has 2 outputs each time while the
other just one
- 3st: The single one starts a puppet wallet loop to distract the
researcher, and making him waste time following this loop as time goes by
```

Can you identify any exchange attribution?

Confidence: medium-high

(This answer is based off the intel we have and some basic research, since i did not finish following the loop and since i didnt see it myself instead of high its medium-high rating).

The WannaCry ransomware attack utilized **three hardcoded Bitcoin addresses** to receive ransom payments, as the malware lacked the capability to generate unique wallet identifiers for individual victims. This design flaw made it difficult for attackers to match specific payments to decryption keys, leading experts to speculate that the perpetrators may not have intended to decrypt files or were unskilled in operational security.

Bitcoin Addresses Used

The three specific wallet addresses hardcoded into the malware, which are publicly visible on the blockchain, were:

- **13AM4VW2dhxYgXeQepoHkHSQuy6NgaEb94**
- **12t9YDPgwueZ9NyMgw519p7AA8isjr6SMw**
- **115p7UMMngoj1pMvkpHijcRdfJNXj6LrLn**

Exchange and Laundering Attempts

Analysis of the blockchain revealed that funds sent to these addresses were later moved in an attempt to launder the cryptocurrency. The attackers attempted to convert the Bitcoin into **Monero** via the exchange network **Shapeshift.io** to obscure the transaction trail due to Monero's higher level of anonymity. This laundering attempt was **thwarted** by Shapeshift, which blacklisted all known addresses associated with the WannaCry attackers to prevent the conversion of the tainted funds.

Attribution Implications

The use of only three fixed addresses, rather than unique addresses per victim, suggests a lack of sophistication among the threat actors. Because the payment addresses were shared among all victims, it was unclear how the attackers would know which specific computers to decrypt after payment. This ambiguity likely contributed to a very low success rate for the ransomware, with only a small fraction of the estimated 200,000 victims actually paying the demanded $300 or $600 ransoms.

What does the transaction pattern tell you about the operator?

Confidence: medium

The dual-output split architecture is deliberate and coordinated — all three wallets executed the same pattern within minutes of each other on Aug 3, 2017.

This level of coordination across three independent wallets in a narrow time window indicates automated tooling or a dedicated financial operations team, not manual execution.

The puppet loop is a deliberate investigator countermeasure — it consumes analyst time while real funds move through the two-output branch.

This is a known professional laundering technique.

Contradiction: Despite sophisticated laundering architecture, the hardcoded wallet addresses are a fundamental OPSEC failure — amateur mistake embedded inside a professional financial operation. This suggests the malware developer and the financial operations team were different people with different skill levels.